

HW2-Yuki

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Question 1

Q1, part A

We know $SST = \sum_{i=1}^N (Y_i - \bar{Y})^2$, $SSR = \sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2$, so we get $R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^N (Y_i - \bar{Y})^2}{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2}$.

We know $\hat{Y}_i = b_0 + b_1 X_i$ and $b_0 = \bar{Y} - b_1 \bar{X}$, so we have $\hat{Y}_i = \bar{Y} - b_1 \bar{X} + b_1 X_i$ and so

$$\begin{aligned} R^2 &= \frac{SSR}{SST} \\ &= \frac{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{\sum_{i=1}^N (\bar{Y} - b_1 \bar{X} + b_1 X_i - \bar{Y})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{\sum_{i=1}^N (b_1 (X_i - \bar{X}))^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{b_1^2 \sum_{i=1}^N (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \end{aligned}$$

For the fitted regression line, we know $b_1 = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$, so we have

$$\begin{aligned} R^2 &= \frac{b_1^2 \sum_{i=1}^N (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \left(\frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2} \right)^2 \frac{\sum_{i=1}^N (X_i - \bar{X})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{(\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}))^2}{\sum_{i=1}^N (X_i - \bar{X})^2 \sum_{i=1}^N (Y_i - \bar{Y})^2} \end{aligned}$$

For the fitted regression line, we know $b_1 = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$ and $b_1 = r_{XY} \times \frac{s_Y}{s_X}$, we have

$$\begin{aligned} r_{XY} &= b_1 \frac{s_X}{s_Y} \\ &= \frac{s_{XY}}{s_X^2} \frac{s_X}{s_Y} \\ &= \frac{s_{XY}}{s_X s_Y} \\ &= \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N (Y_i - \bar{Y})^2}} \end{aligned}$$

and so we get

$$\begin{aligned} (r_{XY})^2 &= \frac{(\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}))^2}{\sum_{i=1}^N (X_i - \bar{X})^2 \sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= R^2 \end{aligned}$$

Q1, part B

We know $\text{corr}(X, e) = \frac{\sum_{i=1}^N (X_i - \bar{X})(e_i - \bar{e})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N (e_i - \bar{e})^2}} = \frac{\sum_{i=1}^N (X_i - \bar{X})e_i}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N e_i^2}}$ since $\bar{e} = 0$. To prove

$\text{corr}(X, e) = 0$, we just need to show the numerator is 0. Thus, using $e_i = Y_i - b_0 - b_1 X_i$ and $\sum_{i=1}^N (X_i - \bar{X}) = 0$ by HW1, we get

$$\begin{aligned} \sum_{i=1}^N (X_i - \bar{X})e_i &= \sum_{i=1}^N (X_i - \bar{X})(Y_i - b_0 - b_1 X_i) \\ &= \sum_{i=1}^N (X_i - \bar{X})Y_i - b_0 \sum_{i=1}^N (X_i - \bar{X}) - b_1 \sum_{i=1}^N (X_i - \bar{X})X_i \\ &= \sum_{i=1}^N (X_i - \bar{X})Y_i - b_1 \sum_{i=1}^N (X_i - \bar{X})X_i \end{aligned}$$

We have $\sum_{i=1}^N (X_i - \bar{X})X_i = \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X} + \bar{X}) = \sum_{i=1}^N (X_i - \bar{X})^2 + \sum_{i=1}^N \bar{X}(X_i - \bar{X}) = \sum_{i=1}^N (X_i - \bar{X})^2 + \bar{X} \sum_{i=1}^N (X_i - \bar{X}) = \sum_{i=1}^N (X_i - \bar{X})^2$ since $\sum_{i=1}^N (X_i - \bar{X}) = 0$ by HW1. We also know $b_1 = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2}$. Thus, we get

$$\begin{aligned}
\sum_{i=1}^N (X_i - \bar{X})e_i &= \sum_{i=1}^N (X_i - \bar{X})Y_i - b_1 \sum_{i=1}^N (X_i - \bar{X})X_i \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - b_1 \sum_{i=1}^N (X_i - \bar{X})^2 \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2} \sum_{i=1}^N (X_i - \bar{X})^2 \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}) \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - \left(\sum_{i=1}^N (X_i - \bar{X})Y_i - \sum_{i=1}^N (X_i - \bar{X})\bar{Y} \right) \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - \left(\sum_{i=1}^N (X_i - \bar{X})Y_i - \bar{Y} \sum_{i=1}^N (X_i - \bar{X}) \right) \\
&= \sum_{i=1}^N (X_i - \bar{X})Y_i - \sum_{i=1}^N (X_i - \bar{X})Y_i \\
&= 0
\end{aligned}$$

and so

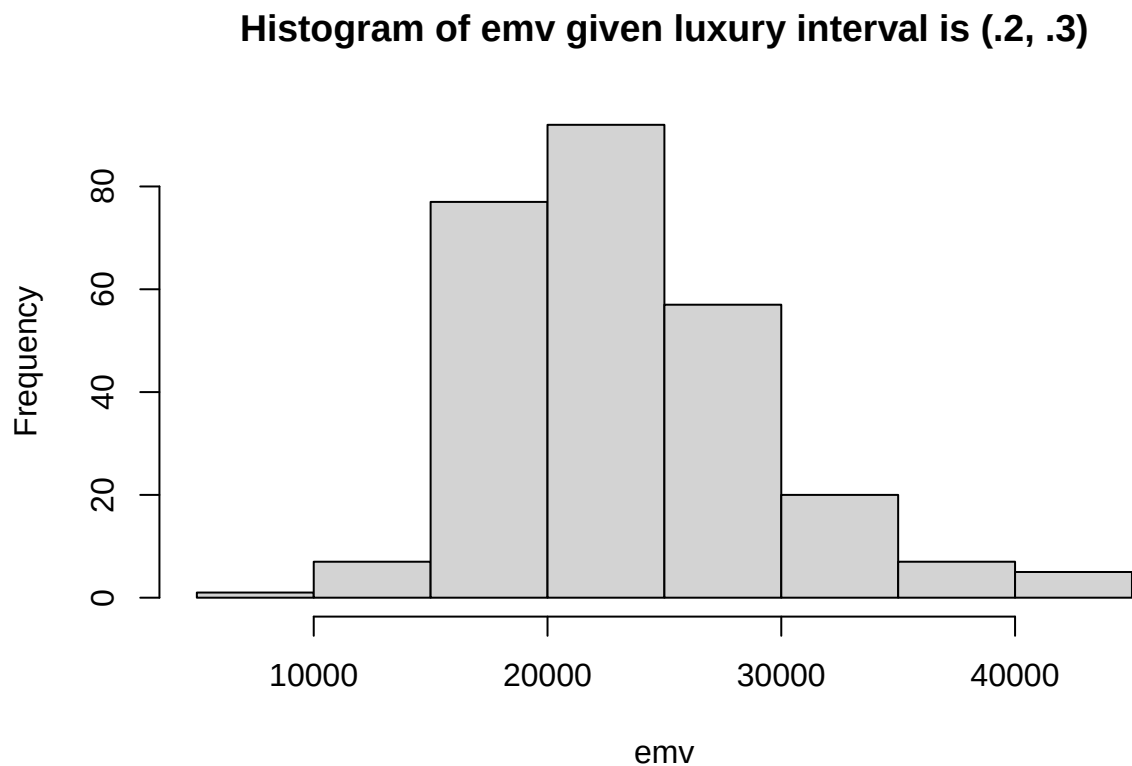
$$\begin{aligned}
corr(X, e) &= \frac{\sum_{i=1}^N (X_i - \bar{X})e_i}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N e_i^2}} \\
&= \frac{0}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N e_i^2}} \\
&= 0
\end{aligned}$$

Question 2 : More on Nearest Neighbor Approaches

Q 2, part A

```
library(DataAnalytics)
data(mvehicles)
cars=mvehicles[mvehicles$bodytype != "Truck",]

luxsubset <- subset(cars, luxury > 0.2 & luxury < 0.3)
hist(luxsubset$emv,
     main = "Histogram of emv given luxury interval is (.2, .3)",
     xlab = "emv")
```



Q 2, part B

```
meanemv <- mean(luxsubset$emv)
meanemv
```

```
## [1] 23376.5
```

```
lwr <- quantile(luxsubset$emv, 0.025)
upr <- quantile(luxsubset$emv, 0.975)
```

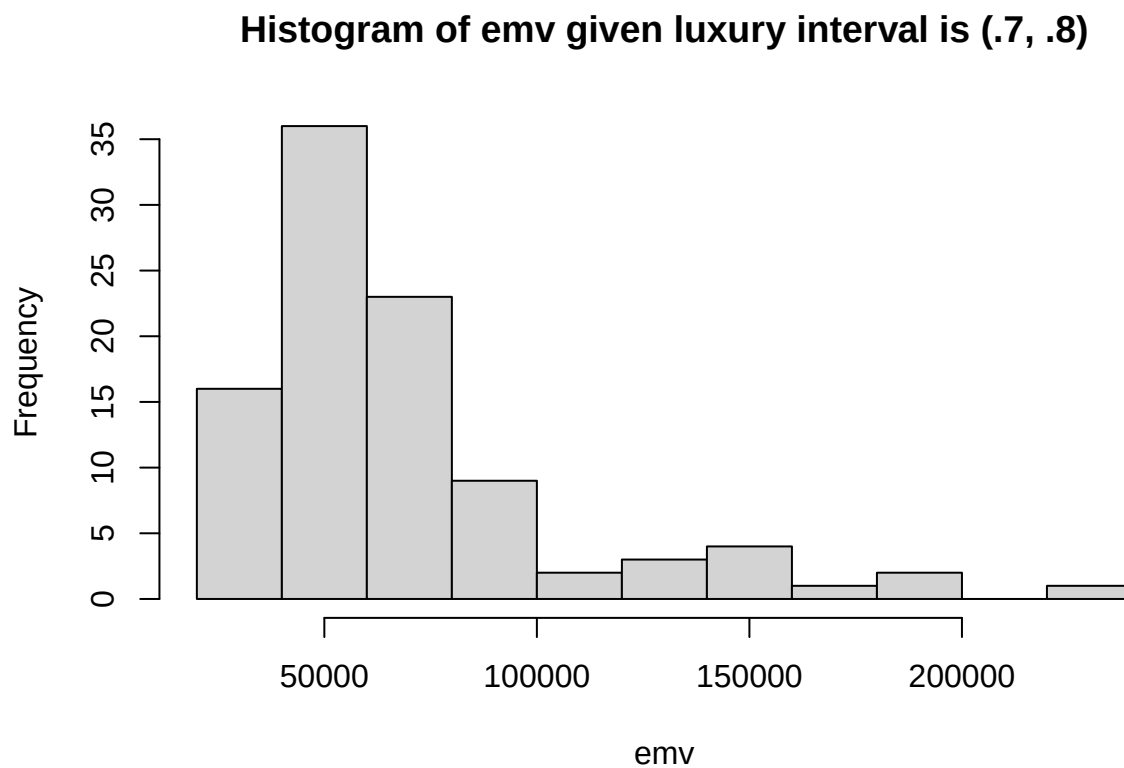
```
predinterval <- c(lwr, upr)
predinterval
```

```
##      2.5%      97.5%
## 14699.09 38632.34
```

The mean of the conditional distribution in part A is 23376.5. The prediction interval that takes up 95% of the data is [14699.09, 38632.34].

Q 2, part C

```
luxsubset2 <- subset(cars, luxury > 0.7 & luxury < 0.8)
hist(luxsubset2$emv,
     main = "Histogram of emv given luxury interval is (.7, .8)",
     xlab = "emv")
```



```
meanemv2 <- mean(luxsubset2$emv)
meanemv2
```

```
## [1] 68101.87
```

```
lwr2 <- quantile(luxsubset2$emv, 0.025)
upr2 <- quantile(luxsubset2$emv, 0.975)
predinterval2 <- c(lwr2, upr2)
predinterval2
```

```
##      2.5%      97.5%
## 34254.33 179179.22
```

The mean of the conditional distribution in part A is 68101.87. The prediction interval that takes up 95% of the data is [34254.33, 179179.22].

By observing these two histograms, we find that the histogram in part A is more symmetric than the histogram in part C. The histogram in part C is skewed to the right. The values of the range of x axis of the histogram in part A is smaller than that in part C. Also, we observe that the range of x axis of the histogram in part A is much narrower than that in part C. The distribution of emv is more concentrated as shown by the histogram in part A. On the contrary, the distribution of emv is more scattered as shown by the histogram in part C. Thus, there are greater variability in the values of emv given luxury is in the interval (.7, .8) than the luxury is in the interval (.2, .3).

By comparing the means of the conditional distribution of emv given luxury in these two intervals, we find that it shows lower average emv value given the luxury is in the interval (.2, .3). On the contrary, the mean of the conditional distribution of emv given the interval is (.7, .8) is higher. Moreover, by observing the prediction intervals that takes up 95% of the data given these two intervals, we find that the prediction interval of emv is narrower given the luxury is in the interval (.2, .3) than the prediction interval of emv given the luxury is in the interval (.7, .8). This indicates that there are more variability of emv values when luxury is higher, which shows that it is not too precise to predict emv based on only luxury variable.

Q 2, part D

Observing the results from part B and part C, it shows that emv is influenced by luxury since we find as the luxury interval increases, the mean values of emv and the prediction intervals increase. However, it is clear that luxury by itself does not provide an accurate prediction of emv because of the wide prediction intervals and obvious variability. Also, there are overlap parts between these two 95% prediction intervals. Thus, using luxury alone cannot predict accurately emv. Specifically, we find that the 95% prediction intervals for emv ranges from 14,699.09 to 38,632.34 given luxury is in the interval (.2, .3), while in the (.7, .8) interval, the range expands to 34,254.33 to 179,179.22. The prediction intervals are too wide to make accurate prediction. Even though the luxury interval is so small, the prediction interval of emv is too wide. It also suggests that as luxury increases, the values of emv become more scattered and spread. This higher variability in the emv values indicate that there are other factors will influence the emv values. Even though the mean emv increases as the luxury increases (from 23,376.5 to 68,101.87), the histograms' broad distribution and lack of tight clustering suggest that this relationship might not be clear and definite. It implies that although luxury and emv may be positively correlated, there are still more variable should be considered to predict emv values more accurately. Additional variables or a more intricate model should be taken into account in order to capture the other influences on emv that cannot be explained by luxury alone.

Question 3 : Optimal Pricing and Elasticities

Q 3, part A

```
data(detergent)

model <- lm(log(q_tide128) ~ log(p_tide128), data = detergent)
summary(model)

##
## Call:
## lm(formula = log(q_tide128) ~ log(p_tide128), data = detergent)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7186 -0.4629 -0.0056  0.4339  2.9980
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.29778    0.13689   97.14  <2e-16 ***
## log(p_tide128) -4.41205    0.06452  -68.38  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7358 on 14743 degrees of freedom
## Multiple R-squared:  0.2408, Adjusted R-squared:  0.2407
## F-statistic: 4676 on 1 and 14743 DF, p-value: < 2.2e-16

elasticity <- coef(summary(model))[2]
elasticity
```

```
## [1] -4.412049
```

```
confint(model, level = 0.9)
```

```
##              5 %      95 %
## (Intercept)  13.072601 13.522963
## log(p_tide128) -4.518186 -4.305912
```

The 90 percent confidence interval for this elasticity is [-4.518186, -4.305912]. The mean price elasticity of demand for 128 oz Tide is -4.412049.

Q 3, part B

```
gross_margin <- 0.25
elas <- -1/gross_margin

conf_interval <- c(confint(model, level = 0.9)[2], confint(model, level = 0.9)[4])
```

```

if (abs(elas) >= conf_interval[1] && elas <= conf_interval[2]) {
  result <- "Fail to reject the null hypothesis"
} else {
  result <- "Reject the null hypothesis"
}

```

```
conf_interval
```

```
## [1] -4.518186 -4.305912
```

```
result
```

```
## [1] "Reject the null hypothesis"
```

Using the inverse elasticity rule, we use the codes above ($\text{elas} < -1/\text{gross_margin}$) to determine the elasticity that is consistent with a 25 percent gross margin is -4.

$H_0 : \text{elasticity} = -4$

$H_A : \text{elasticity} \neq -4$

The 90% confidence interval, which is [-4.518186, -4.305912], does not contain -4, so we should reject the null hypothesis that elasticity = -4. The retailer is not pricing optimally at the 90 percent confidence level.

Question 4

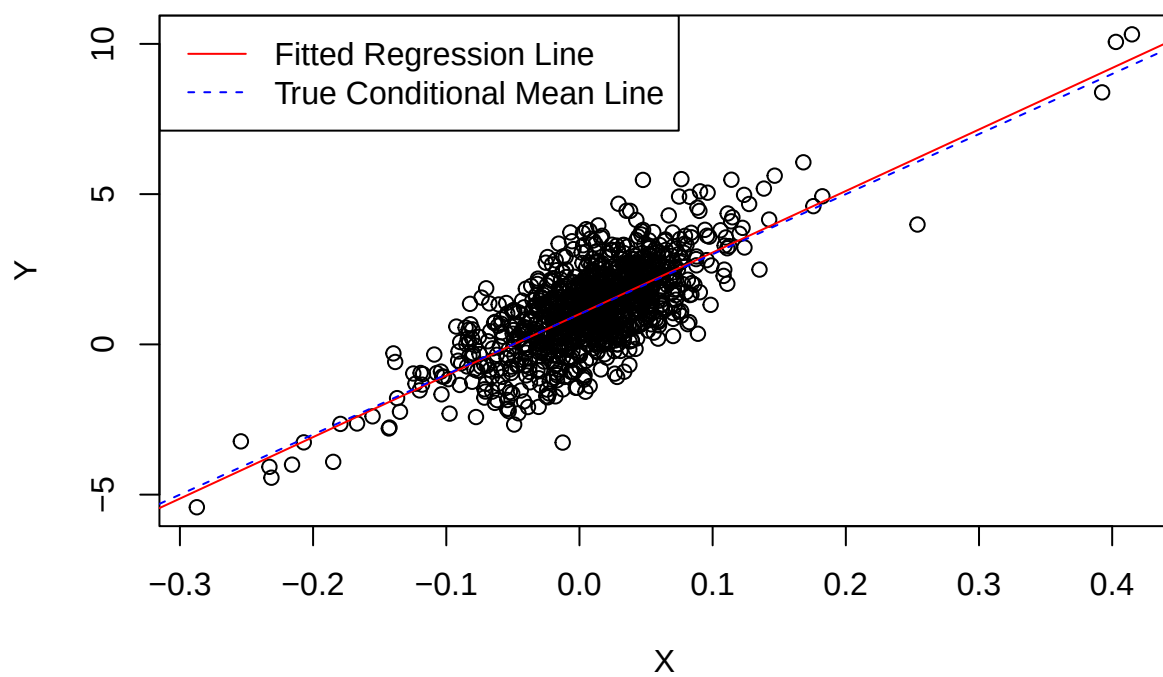
a.

```
regression_model <- function(b0, b1, X, sigma) {  
  error <- rnorm(length(X), mean = 0, sd = sigma)  
  Y <- b0 + b1*X + error  
  return(Y)  
}
```

b.

```
data(marketRf)  
  
b0 = 1  
b1 = 20  
sigma = 1  
  
x_values <- marketRf$vwretd  
y_values <- regression_model(b0, b1, x_values, sigma)  
  
plot(x_values, y_values, main = "Scatterplot of X vs simulated Y",  
      xlab = "X", ylab = "Y",)  
  
fitted_model <- lm(y_values ~ x_values)  
abline(fitted_model, col = "red", lty = 1)  
  
abline(a = b0, b = b1, col = "blue", lty = 2)  
  
legend("topleft", legend = c("Fitted Regression Line", "True Conditional Mean Line"),  
       col = c("red", "blue"), lty=c(1,2))
```

Scatterplot of X vs simulated Y



Question 5

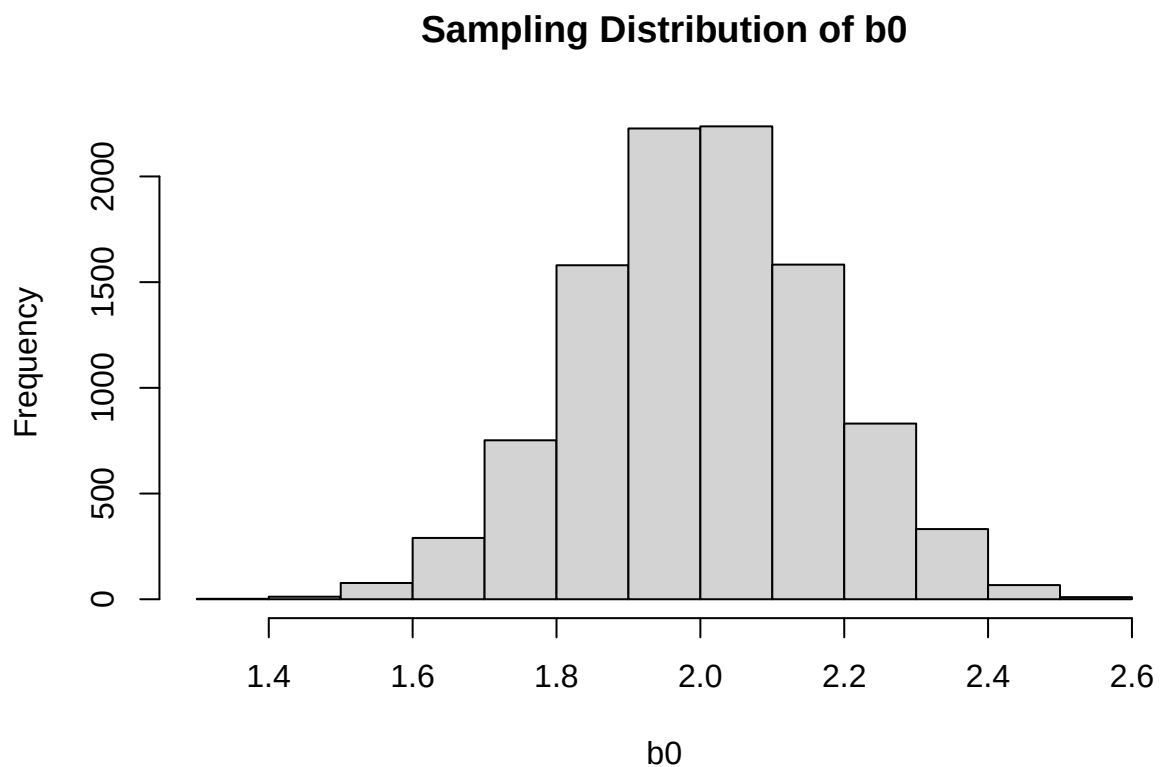
```
beta0 <- 2
beta1 <- 0.6
sigmasquare <- 2
```

a.

```
set.seed(2810510)
N <- 300
sigmanew <- sqrt(sigmasquare)

samplesnum <- 10000
beta0samples <- numeric(samplesnum)
beta1samples <- numeric(samplesnum)
x_uniform <- runif(N, min = 0, max = 1)
for (i in 1:samplesnum){
  y_simulate <- regression_model(beta0, beta1, x_uniform, sigmanew)
  modelnew <- lm(y_simulate ~ x_uniform)
  beta0samples[i] <- coef(modelnew)[1]
  beta1samples[i] <- coef(modelnew)[2]
}

hist(beta0samples, main = "Sampling Distribution of b0",
      xlab = "b0")
```



b.

```
empb1 <- mean(betalsamples)
empb1
```

```
## [1] 0.5967556
```

```
theob1 <- beta1
theob1
```

```
## [1] 0.6
```

The simulated value for $E[b_1]$ is 0.5967556 and the theoretical value is 0.6. These two values are very close.

c.

```
empvarb1 <- var(betalsamples)
empvarb1
```

```
## [1] 0.08460096
```

```
xbar <- mean(x_uniform)
deno <- sum((x_uniform - xbar)^2)
theovarb1 <- sigmanew^2 / deno
theovarb1
```

```
## [1] 0.08348097
```

The simulated value for $Var(b_1)$ is 0.08460096. The theoretical value is 0.08348097. These two values are close.

Question 6

- a. The standard error is the estimated standard deviation of the sampling distribution. It is used to estimate the variability of the estimator. It indicates how much the sample statistic is expected to vary from sample to sample if we were to repeatedly take samples from the same population. Standard deviation and standard error both measure variability. However, the standard error calculates the variability across several samples of a population to estimate the precision of a sample statistic when making inferences about a population, whereas the standard deviation characterizes the variability within a set of data or measurements.
- b. The sampling error is defined as the differences between the samples and the true population. It happens when an analyst fails to choose a sample that accurately reflects the total population of data. The bias brought on by sampling is known as sampling error. The magnitude of sampling error is measured by the standard error. When we take many random samples from the same population, it measures the expected variability of a sample statistic. Greater variability across sample statistics from various samples is indicated by a higher standard error, which suggests a larger sampling error. This implies that the sample statistic's ability to accurately estimate the population parameter may be compromised. Less variability across sample statistics is shown by a lower standard error, which means that the sample statistic is a more accurate estimator of the population parameter and the sampling error is smaller.
- c. Based on the meaning of standard error, we can tell Steven that if the standard error is smaller, it indicates that the parameter estimate is more precise, while a larger standard error indicates more variability. The standard error can be used to calculate the test statistics, which could help determine how closely the distribution of Steven's data matches the distribution expected under the null hypothesis of the statistical test Steven is using. Using standard error could help construct the confidence interval. It provides information about the range of values of the parameter. This would help Steven to have better ideas about what the value of true parameter is since it is likely that the confidence interval provides a range of values that contain the true value of the parameter. The standard error also can be used to get the p-value and decide if we should reject the null hypothesis or not.
- d. Test statistic is a calculated number derived from the data that indicates the degree to which the sample differ from the distribution expected under null hypothesis. The greater test statistic is, typically the stronger evidence against the null hypothesis is. For the p-value, it is the probability of observing a value of the test statistics farther out in the tail than the observed test statistic value as taught in class. Assuming the null hypothesis is true, the p-value indicates the likelihood that Xingua will see a test statistic that is as extreme as or more extreme than the one that was obtained. Thus, if the p-value is smaller than the significance level, it indicates that there is strong evidence that could reject the null hypothesis.